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Roll No

CS-7005(2)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Data Science and Big Data

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Give a detailed note on features and limitations of Python programming. 7
b) How analytical tools have evolved from graphical user interfaces to point solutions to data visualization tools? 7
2. a) Discuss why is big data analytics important. 7
b) For each of the following scenarios, state which one is more likely to lead to the rejection of the null hypothesis
i) A one-tailed or two-tailed test
ii) .05 or .01 level of significance
iii) A sample size of $n = 144$ or $n = 444$
7
3. A teacher draws a sample of 5, 12-year-old children from the school's population and records their heights as follows:
{124, 124, 128, 130, 127}
Assume that the heights have a normal distribution where both μ and σ are unknown. Calculate a two-tailed 95% confidence interval for the mean height of 12-year-olds. 14

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4. Write short notes on any two: 14
- a) Logistic regression
 - b) Back propagation algorithm
 - c) Issues in machine learning
5. a) Explain about the Information Retrieval. 7
- b) Write the advantages and disadvantages for classic models which are used in IR and discriminate their techniques. 7
6. a) Briefly explain Web search architectures. 7
- b) Write the formal characterization of IR models. 7
7. a) Describe the working of Map reduce with an relevant example. 7
- b) Discuss the points to be considered while designing a file system in Map-reduce. 7
8. a) The product moment correlation coefficient between the random variables W and X is 0.71 and between the random variables Y and Z is -0.05 . For each of these pairs of variables, sketch a scatter diagram which might represent the results which gave the correlation coefficients. 7
- b) The values of independent variable X and dependent value Y are given below: 7

X	Y
0	2
1	3
2	5
3	4
4	6

Find the least square regression line $y = ax + b$. Estimate the value of y when x is 10.
